

1 五則演算 解答

$5 + 8 = 13$	$9 - 4 = 5$	$2 * 3 = 6$	$6 / 2 = 3$	$5 \% 2 = 1$
$-9 + 4 = -5$	$3 - 10 = -7$	$-4 * 5 = -20$	$-10 / 5 = -2$	$10 \% 3 = 1$
$15 + 27 = 42$	$50 - 18 = 32$	$12 * 4 = 48$	$48 / 4 = 12$	$25 \% 4 = 1$
$-50 + 12 = -38$	$20 - 85 = -65$	$15 * -3 = -45$	$100 / -25 = -4$	$100 \% 7 = 2$
$1.2 + 3.8 = 5$	$4.5 - 1.2 = 3.3$	$0.5 * 6 = 3$	$4.8 / 2 = 2.4$	$50 \% 6 = 2$
$-4.5 + 1.2 = -3.3$	$0.8 - 3.5 = -2.7$	$-1.2 * 4 = -4.8$	$-7.2 / 0.9 = -8$	$80 \% 9 = 8$
$125 + 380 = 505$	$500 - 125 = 375$	$25 * 12 = 300$	$144 / 12 = 12$	$123 \% 10 = 3$
$-600 + 250 = -350$	$100 - 456 = -356$	$14 * -10 = -140$	$-117 / 9 = -13$	$250 \% 15 = 10$
$0.45 + 0.55 = 1$	$1.25 - 0.75 = 0.5$	$0.8 * 0.5 = 0.4$	$0.75 / 0.25 = 3$	$1000 \% 3 = 1$
$-12.5 + 7.5 = -5$	$1000 - 1500 = -500$	$-15 * 1.2 = -18$	$102 / -3 = -34$	$1024 \% 10 = 4$

考え方 加減乗除は符号規則に注意し、%は「割った余り」を求める。小数は必要なら 10 倍して整数化して考える。

途中式（代表例）

$$-4.5 + 1.2 = -(4.5 - 1.2) = -3.3$$
$$-7.2 / 0.9 = -72 / 9 = -8$$
$$250 \% 15 = 250 - 15 \times 16 = 10$$

2 ルートと乗数の計算 解答

$2^2 = 4$	$2^2 * 2^1 = 8$	$\sqrt{4} = 2$	$\sqrt{4} * \sqrt{9} = 6$
$3^2 = 9$	$3^3 / 3^1 = 9$	$\sqrt{9} = 3$	$\sqrt{2} * \sqrt{8} = 4$
$2^4 = 16$	$2^2 * 2^2 = 16$	$\sqrt{16} = 4$	$\sqrt{12} / \sqrt{3} = 2$
$5^2 = 25$	$10^2 / 10^1 = 10$	$\sqrt{25} = 5$	$\sqrt{3} * \sqrt{12} = 6$
$4^3 = 64$	$2^3 * 2^2 = 32$	$\sqrt{49} = 7$	$\sqrt{18} / \sqrt{2} = 3$
$6^2 = 36$	$5^3 / 5^2 = 5$	$\sqrt{64} = 8$	$\sqrt{2} * \sqrt{32} = 8$
$10^2 = 100$	$2^2 * 2^4 = 64$	$\sqrt{100} = 10$	$\sqrt{50} / \sqrt{2} = 5$
$12^2 = 144$	$3^4 / 3^2 = 9$	$\sqrt{144} = 12$	$\sqrt{72} / \sqrt{2} = 6$

考え方 指数法則 $a^m a^n = a^{m+n}$, $a^m / a^n = a^{m-n}$ 、根号の性質 $\sqrt{a} \sqrt{b} = \sqrt{ab}$ を使う。

途中式（代表例）

$$3^3 / 3^1 = 3^{3-1} = 3^2 = 9$$
$$\sqrt{12} / \sqrt{3} = \sqrt{12 / 3} = \sqrt{4} = 2$$
$$\sqrt{2} \cdot \sqrt{32} = \sqrt{64} = 8$$

3 分数の計算 解答

$\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$	$\frac{4}{5} - \frac{1}{5} = \frac{3}{5}$	$\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$	$\frac{2}{5} \div 2 = \frac{1}{5}$
$-\frac{2}{3} + \frac{1}{3} = -\frac{1}{3}$	$\frac{1}{4} - \frac{3}{4} = -\frac{1}{2}$	$-\frac{1}{2} \times \frac{1}{4} = -\frac{1}{8}$	$\frac{1}{5} \div (-2) = -\frac{1}{10}$
$\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$	$\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$	$\frac{2}{3} \times \frac{3}{5} = \frac{2}{5}$	$\frac{3}{4} \div 2 = \frac{3}{8}$
$-\frac{1}{2} + \frac{1}{6} = -\frac{1}{3}$	$\frac{1}{6} - \frac{1}{2} = -\frac{1}{3}$	$\frac{2}{5} \times \left(-\frac{1}{4}\right) = -\frac{1}{10}$	$-\frac{3}{4} \div 3 = -\frac{1}{4}$
$\frac{3}{4} + \frac{1}{6} = \frac{11}{12}$	$\frac{5}{6} - \frac{1}{4} = \frac{7}{12}$	$\frac{5}{8} \times \frac{4}{15} = \frac{1}{6}$	$\frac{2}{3} \div \frac{1}{2} = \frac{4}{3}$
$-\frac{1}{2} - \frac{1}{3} - \frac{1}{6} = -1$	$\frac{1}{10} - \frac{4}{15} = -\frac{1}{6}$	$-\frac{15}{16} \times \frac{4}{5} = -\frac{3}{4}$	$10 \div \left(-\frac{5}{2}\right) = -4$

考え方 加減は通分、乗算は分子同士・分母同士、除算は逆数をかける。最後に約分する。

途中式（代表例）

$$\begin{aligned}\frac{3}{4} + \frac{1}{6} &= \frac{9}{12} + \frac{2}{12} = \frac{11}{12} \\ \frac{2}{3} \div \frac{1}{2} &= \frac{2}{3} \times \frac{2}{1} = \frac{4}{3} \\ -\frac{1}{2} - \frac{1}{3} - \frac{1}{6} &= -\frac{3}{6} - \frac{2}{6} - \frac{1}{6} = -1\end{aligned}$$

4 方程式の計算 解答

$x + 5 = 8 \Rightarrow x = 3$	$3x = 12 \Rightarrow x = 4$	$x + 10 = 4 \Rightarrow x = -6$
$2x - 3 = 7 \Rightarrow x = 5$	$15 - x = 20 \Rightarrow x = -5$	$4x + 16 = 0 \Rightarrow x = -4$
$3x + 4 = x + 10 \Rightarrow x = 3$	$2x - 5 = 5x + 7 \Rightarrow x = -4$	$4x - 8 = 6x + 2 \Rightarrow x = -5$
$2(x - 3) = 4 \Rightarrow x = 5$	$3(x + 4) = -6 \Rightarrow x = -6$	$5(x + 1) = 2x - 4 \Rightarrow x = -3$
$0.2x = 4 \Rightarrow x = 20$	$\frac{x}{3} - 2 = -4 \Rightarrow x = -6$	$0.5x + 3 = 1 \Rightarrow x = -4$

考え方 移項で符号反転、係数は逆演算で消す。かっこは先に展開。

途中式（代表例）

$$\begin{aligned}2x - 5 &= 5x + 7 \Rightarrow -5 - 7 = 5x - 2x \Rightarrow -12 = 3x \Rightarrow x = -4 \\ 3(x + 4) &= -6 \Rightarrow x + 4 = -2 \Rightarrow x = -6 \\ \frac{x}{3} - 2 &= -4 \Rightarrow \frac{x}{3} = -2 \Rightarrow x = -6\end{aligned}$$

5 式の展開 解答

$$5(x+3)=5x+15$$
$$-2(4x-7)=-8x+14$$
$$(x+8)^2=x^2+16x+64$$
$$(x+12)(x-12)=x^2-144$$
$$(3x-1)(2x+5)=6x^2+13x-5$$

$$(x+2)(x+6)=x^2+8x+12$$
$$(x-3)(x+5)=x^2+2x-15$$
$$(x-9)^2=x^2-18x+81$$
$$(2x+3)(x+2)=2x^2+7x+6$$
$$(4x-3)^2=16x^2-24x+9$$

考え方 分配法則で全項にかける。公式 $(a \pm b)^2 = a^2 \pm 2ab + b^2$ 、 $(a + b)(a - b) = a^2 - b^2$ を活用。

途中式（代表例）

$$(x+2)(x+6)=x^2+6x+2x+12=x^2+8x+12$$
$$(x-9)^2=x^2-18x+81$$
$$(3x-1)(2x+5)=6x^2+15x-2x-5=6x^2+13x-5$$

6 因数分解 解答

$$2x+10=2(x+5)$$
$$x^2-2x-8=(x-4)(x+2)$$
$$x^2+6x+9=(x+3)^2$$
$$x^2+x-20=(x+5)(x-4)$$
$$x^2-81=(x-9)(x+9)$$

$$x^2+7x+10=(x+5)(x+2)$$
$$x^2-5x+4=(x-1)(x-4)$$
$$x^2-25=(x-5)(x+5)$$
$$x^2+10x+25=(x+5)^2$$
$$x^2+13x+42=(x+6)(x+7)$$

考え方 共通因数でくくるか、和と積の公式を使う。二次式は「和が中項、積が定数項」。

途中式（代表例）

$$x^2-2x-8=(x-4)(x+2) \quad (-4+2=-2, \ -4 \cdot 2=-8)$$
$$x^2+13x+42=(x+6)(x+7)$$
$$x^2-81=x^2-9^2=(x-9)(x+9)$$

7 文字と式 解答

- 1. $5a + 3b = c$
- 2. $1000 - 3x = y$
- 3. $vt = d$
- 4. $100 - xt = y$
- 5. $1.1(2x + 5y) = z$
- 6. $\frac{a + b + c + d}{4} = m$
- 7. $x - vt = p$
- 8. $10a \left(1 - \frac{x}{100}\right) = b$

考え方 文章を「合計」「差」「速さ × 時間」「平均」に分解し、未知量を式でつなぐ。

途中式（代表例）

- 2. 支払い 1000 - 代金 $3x =$ おつり y
- 6. 平均 $= \frac{\text{合計}}{4} = \frac{a + b + c + d}{4} = m$
- 8. 1 個の割引後価格 $= a \left(1 - \frac{x}{100}\right)$, 10 個で $10a \left(1 - \frac{x}{100}\right) = b$